

BGL Data & AI Team: Coding Challenge

We are interested to see your coding, problem solving and collaboration style. We would like you to build the following system.

We recommend applicants to use Java, TypeScript or Python, but you could choose whatever you are comfortable with.

What to include in your submission?

- A link to a GitHub repository or a zip file of your code. If you provide a zip, make sure that it contains a GIT repository, so we can see the **commit history**.
- Appropriate **documentation** (README file) - it should be clear about how to setup, run, and test your code.
- Appropriate **unit and/or integration tests**.

What are we looking for in a good solution?

- Readability, design, clean and maintainability of code
- A test-driven development (TDD) approach
- Documentation skills

Introduction

An online grocery shop is in the process of rewriting its products management and ordering system.

Part 1 – Products

Code	Name	Price
CE	Cheese	\$5.95
HM	Ham	\$7.95
SS	Soy Sauce	\$11.95

Products are regularly updated, so the system should support for creating, retrieving, deleting, and modifying products.

Part 2 – Packaging Options

The shop would like to offer packaging options that sell products in bundles, and charge customers on a discounted price.

Code	Packaging Options
CE	3 for \$14.95 5 for \$20.95
HM	2 for \$13.95 5 for \$29.95 8 for \$40.95

Part 3 - Order

A customer order consists of pairs of product codes and counts. For example, 10 x CE, 14 x HM, and 3 x SS. The system should return the cost and packaging breakdown for each requested product.

To save shipping cost, **the order should contain the minimal number of packages.**

Example

Given an order of 10 CE, 14 HM, and 3 SS, the correct breakdown is:

- 10 CE for \$41.90
 - 2 packages of 5 items (\$20.95 each)
- 14 HM for \$78.85
 - 1 package of 8 items (\$40.95 each)
 - 1 package of 5 items (\$29.95 each)
 - 1 package of 1 item (\$7.95 each)
- 3 SS for \$35.85
 - 3 packages of 1 item (\$11.95 each)

Tips

We value work-life balance, and do not want you to lose a weekend trying to solve this problem. Only spend enough time required to produce a clean, testable, and maintainable solution to the stated problem. You should focus on delivering only a back-end OR front-end implementation. Keep it simple.